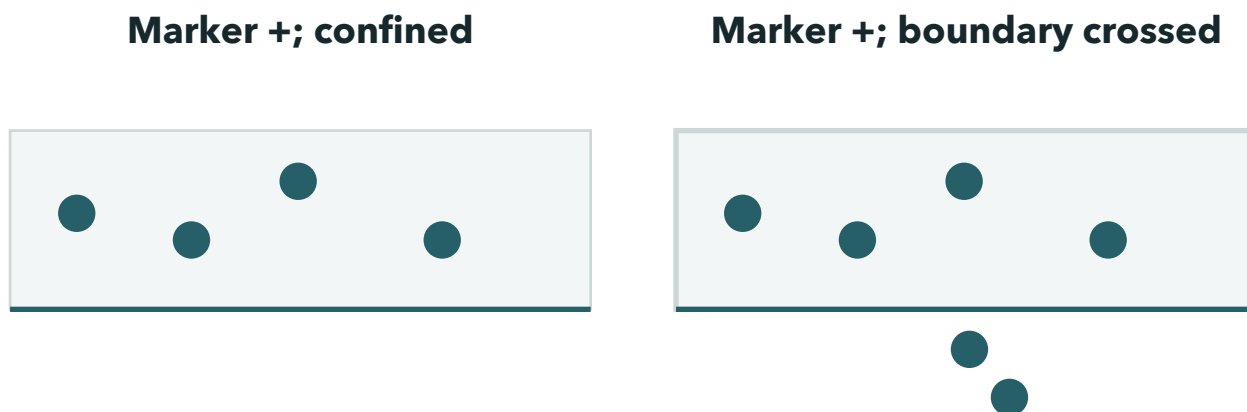


Histopathology: architecture and evidence

Part 1 | A stain and an architectural pattern answer different questions

Cell lineage, tissue invasion and cause require distinct evidence.



Boundary relationship determines invasion evidence

Epithelial-marker positivity is illustrated on both sides of a boundary; invasion requires the architectural relationship, not the colour alone.

Read in layers

Begin with specimen site and architecture, then cell morphology and appropriate ancillary tests. Cytokeratin supports an epithelial phenotype in context; it does not by itself prove malignancy or invasion. The tissue relationship and the full marker pattern remain important.

Granuloma is a pattern, not a single cause

Epithelioid histiocytes and giant cells reflect a coordinated tissue response.

Evidence

Feature	What it adds	Limitation
Granulomatous morphology	A reaction pattern	Several causes possible
Necrosis	Another descriptive feature	Not universally organism-specific
Organism or foreign-material evidence	Potential causal clue	Sampling and assay limits remain

A causation question

Non-necrotising granulomas can support sarcoidosis in the appropriate clinical setting after alternatives are assessed. Their presence alone is not equivalent to demonstrating a particular infection or excluding every infection. Avoid replacing morphology with an etiological label before reviewing the rest of the evidence.

A marker panel constrains lineage

One positive stain is weaker than concordant morphology, controls and a coherent panel.

Interpret immunohistochemistry

- Check positive and negative controls
↓
- Identify the stained cell population
↓
- Assess localisation and distribution
↓
- Integrate the panel with morphology

An original comparison

A pan-cytokeratin-positive lesion still needs architectural assessment and other evidence before an invasive carcinoma label is assigned. Staining normal adjacent epithelium also provides context. A technically convincing positive reaction answers antigen expression; it does not automatically answer every biological question about the lesion.

Margins describe the examined specimen

A margin conclusion depends on the submitted tissue and its orientation.

Report term

Feature	Meaning	Useful distinction
Positive margin	Tumour reaches examined edge	Residual disease may remain
Negative margin	No tumour at examined edge	Not a guarantee about unsampled tissue
Grade	Cellular/tissue differentiation	Not the same as anatomical stage

Specimen context

State the site, procedure and sampling orientation when requesting pathology. A fragmented unoriented sample can constrain interpretation of its edges. An accurate histological diagnosis and an uncertain excision margin can coexist; they are separate report questions.

Keep the evidence chain visible

A diagnosis becomes more defensible when every conclusion has the appropriate supporting observation.

Conclusion

Feature	Required evidence level
Reaction pattern	Architecture and morphology
Cell lineage	Morphology plus validated panel
Invasion	Relationship across tissue boundary
Specific infection	Organism-directed evidence in context

Examination approach

Ask whether the option is a description, a mechanism, a cause or a stage. Then compare it with the evidence actually supplied. A question can intentionally give enough information for the first conclusion while leaving the others unresolved.